Mapping Africa's Endangered Archaeological Sites and Monuments (MAEASaM): Administrative model

Arches Resource Model Graph and Reference Data (Thesaurus)

Version 2

Date: 2026-05-21

Coordinator/s: Renier van der Merwe

Quality control

Name

Contents

Document history
20260521 - version

Schema Attributes and Definitions

All resource model metadata templates in the MAEASaM database, including Remote Sensing Resource Model, are described using the following shared attributes:

Group Name: In Arches, this corresponds to the Semantic Node. It represents the parent category under which related metadata elements are organised.

Node Name: Also referred to as a metadata element. This is the primary descriptive field used to document a Remote Sensing record.

CIDOC CRM Class: Indicates the equivalent class in the CIDOC CRM ontology. This mapping supports semantic interoperability.

CIDOC CRM Property: Indicates the equivalent property in CIDOC CRM ontology. This mapping supports semantic interoperability.

Connector Node: Specifies whether the node establishes a relationship with another resource model. Values are "yes" or "no".

Connected To: If the Connector Node value is "yes", this field specifies the name of the related Resource Model.

Data Type: Defines the type of data entered in Arches. Common data types include: string, date, concept list, and resource-instance-list.

Necessity: Indicates whether the node is required when describing a record. Possible values are:

"mandatory"

"strongly suggested"

Cardinality: Specifies how many values a node can contain. In the MAEASaM Arches context, four options are used:

1 - Only one value allowed

0..1 - Optional, maximum one value

1..n - At least one value required, multiple allowed

0..n - Optional, multiple values allowed

Scope Note: Provides a definition and conceptual explanation of the node.

Protocol for Entry: Explains how information should be entered into this node, including formatting rules, conventions, or controlled vocabulary guidance.

Section 1: Overview

1.1 Objectives and Scope

The function of the administration model is to provide the administration boundaries for each of the participating partner countries. However, the MAEASaM project acknowledges that a number of partner countries are involved in border disputes or have contested territorial delimitations. As such, the MAEASaM project has a neutral stance regarding all territorial disputes and delimitations, and, as such, the project did not determine any of the territorial delimitations that are used in the administrative model.

In line with the project's stance of neutrality, all administrative borders used are those that are officially recognised by the United Nations and supplied by the countries themselves to the United Nations. Therefore, the administration files used in the creation of the administration model were obtained from the Humanitarian Data Exchange (HUM) (https://data.humdata.org/). Although changes to the attributes tables of each country were made to bring the files in line with the structure of the administrative model (only the project identifier and the name of the administrative level were added, with all other modifications being limited to data organisation), no modifications, additions or removals were made related to the geographical organisation or territorial extent of each file. Therefore, any disputes or disagreements related to the geographical extent of the files used in the model should be directed to the Humanitarian Data Exchange.

Geographic extent: The administrative model includes the territorial boundaries of the 16 African countries, with the following countries having been added to the model: Benin, Botswana, Democratic Republic of the Congo, Ethiopia, Ghana, Guinea-Bissau, Kenya, Mali, Mozambique, Nigeria, Senegal, Sudan, Tanzania, The Gambia, Togo and Zimbabwe.

1.2 History of the creation of the Administration model

The HUM data set provides the administrative files of each country in one or more file formats. For the countries in which the project is operating, the file formats available were shapefile, GeoJSON and Geopackage (except for Botswana and Tanzania). It was decided to download the GeoJSON files for each country for the following reasons: Firstly, each of the project partner countries in the HUM database had a GeoJSON file present. Secondly, as standalone files for each of the administration levels, they are easier to edit and organise in order to bring the file attributes in line with the Administrative Resource Model structure. All files used were downloaded on 2026-01-12 and represented the most up-to-date version available on the database at the time of download.

The admin levels downloaded were from level 0 to level 2. It was decided not to include level 3 due to the size of the level 3 admin boundaries, as well as the limitation in the HUM dataset, in which most countries do not have level 3 data available. After downloading, each of the 16 countries has three GeoJSON files associated with it, one file each for levels 0, 1, and 2. These files were then added to QGIS per level. This resulted in 16 layers for every level, which represented each of the project countries. At this stage, the MAEASaM ID for each country was added using the following QGIS expression:

'ADMN-[country ISO 3166-1 alpha-3 number]-' || lpad(@row_number , 8, '0') for the level 0

and

'ADMN-[country ISO 3166-1 alpha-3 number]-' || lpad(@row_number + 1, 8, '0') for level 1

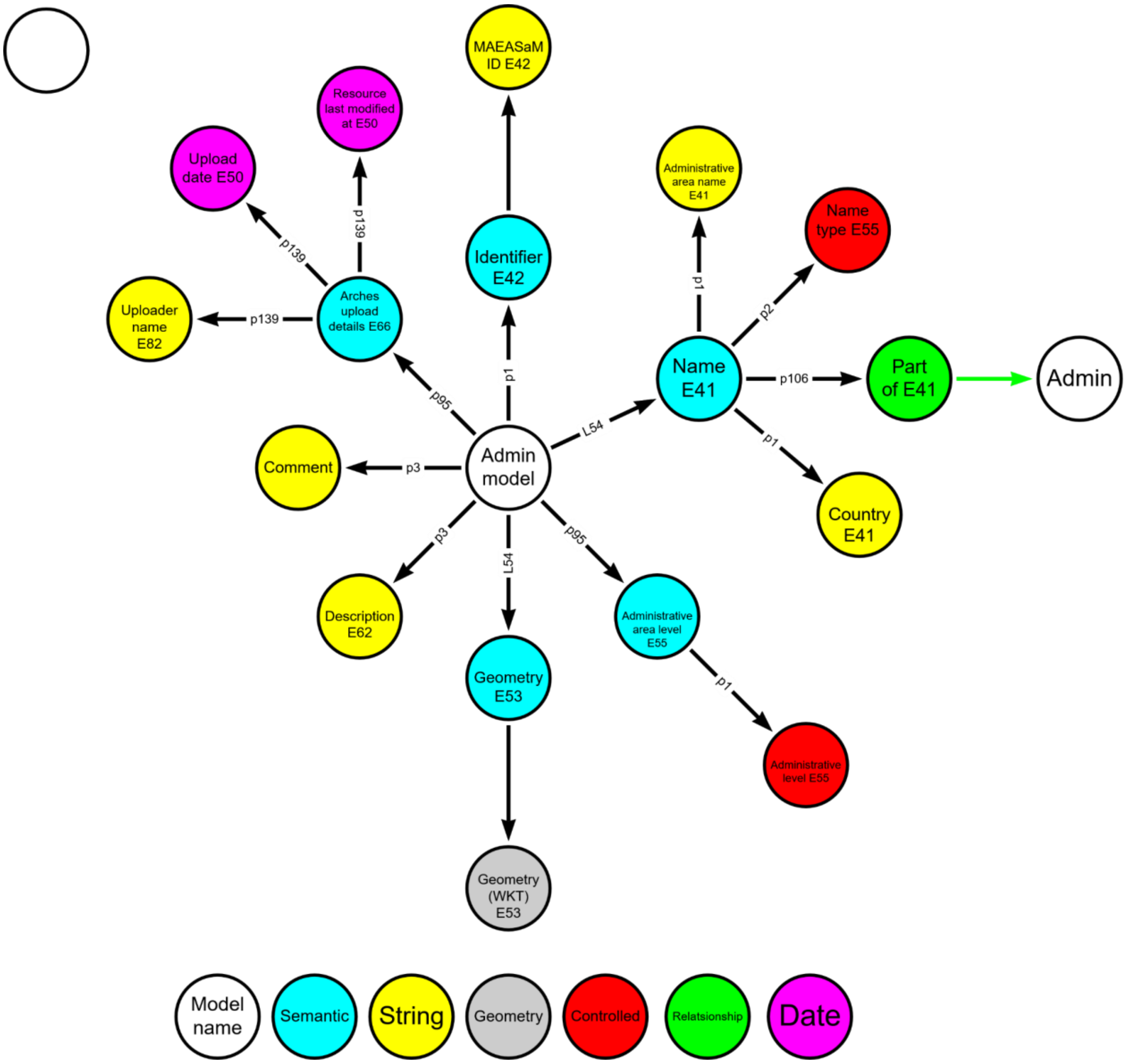
and

'ADMN-[country ISO 3166-1 alpha-3 number]-' || lpad(@row_number + [last number of level 1], 8, '0') for level 2

This expression takes the row number of each entry, and then adds the number indicated in the expression, after which the suffix is added. For example, for Zimbabwe (ZWE), the first number created would be ADMN-ZWE-00000001. Once the MAEASaM ID was added, the name of the administration level was also added (this varied from country to country and was thus added per country and per level). Once the MAEASaM ID and admin level name were added for each level and country, the different layers were then merged based on the admin level for each country. At the end of this process, three GeoJSON files were created containing the same administrative layer for each country. After a layer for each admin level was created, the editing of the attributes table was started. This was done to ensure that all changes and modifications would be applied to all countries equally, to reduce the risk of human error. Attribute headers that are to be used in the Admin resource model (such as adm0_name) had the attribute name changed to match the structure required. Additional fields (validation date and version) were kept, even though they are not used in the administration resource model. This was done for data quality and validation purposes. All additional attributes were deleted, and the remaining attributes that would apply to all the entries were added in bulk via a QGIS expression to further reduce any potential human error. Once this process was finished and all the modifications to the attribute tables were made, each country was exported as a geopackage that contained the level 0, 1 and 2 administration levels.

2. Resource model graph

2.1 Structure of the Administrative Model Resource Graph



A graphical representation of the graph structure used in the Administrative resource model using the CIDOC-CRM version 6.2 ontology (https://cidoc-crm.org/Version/version-6.2). Each colour represents a different data node used within the Arches database, the CIDOC class (E) is used and how that node relates to other nodes via the CIDOC property (p).

2.2 Administrative Resource Model Nodes and CIDOC-CRM Structure

Semantic Node: Identifier

Node Name: MAEASaM ID
CIDOC CRM Class: E42 Identifier
CIDOC CRM Property: p1 is identified by
Connector Node: No
Connected to: N/A
Data type: String
Name of concepts collection: N/A
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)
This node captures the value of the MAEASaM ID for each administrative entry and uses the E42 identifier value in the CIDOC-CRM (v6.2) ontology and is linked with the p1 is identified by property. The ID is an alphanumeric identifier separated into three sections, divided by a dash (-). The first section of the id has the name of the model to which it belongs, and for this model it is represented by "ADMIN". This is followed by the International Organization for Standardization (ISO 3166) 3 digit country code for each country in the admin model (see appendix A). The numerical part follows sequentially, starting from 1. The design of the ID structure uses a fixed character size, and thus the first number will be preceded by 8 zeros, to create a numerical number consisting of 8 characters. As the sequential number increases, for every 10, 100, 1000 etc, the corresponding zero will be removed to ensure the character length remains 8 characters.

Example:
ADMIN-ZWE-00000001
ADMIN-GMB-00000010
ADMIN-KEN-00000100
ADMIN-SEN-00001000

Protocol for entry:

- Allow only one ID per entry.
- Determine the ID value based on the relevant entry.

Semantic Node: Name

Node Name: Administrative area name
CIDOC CRM Class: E41 Appellation
CIDOC CRM Property: p1 is identified by
Connector Node: No
Connected to: N/A
Data type: String
Name of concepts collection: N/A
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)
This node provides the official name of the administrative area and uses the E41 class and p1 property. The names are derived from the HUM dataset and have not been altered in the resource model. Additional names can be added to capture regional variations or historical versions of an administrative area name.

Example:
Mali

Protocol for entry:
- Determine which level of the admin is being used.
- Select the administrative unit that needs to be added, and obtain the name from the downloaded HUM dataset.
- Add the name HUM dataset name version to this node.

Node Name: Name type
CIDOC CRM Class: E55 Type
CIDOC CRM Property: p2 has type
Connector Node: No
Connected to: N/A
Data type: Controlled-list
Name of concepts collection: Name type
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)
This node indicates what type of name has been captured in the above node and uses E55 class with the p2 property.

Example:
Official

Historical
Local

Protocol for entry:

- Determine what name has been captured in the previous node.
- Select the correct term from the Name Type vocabulary that matches the type of name that has been added.

Node Name: Part of
CIDOC CRM Class: E41 Appellation
CIDOC CRM Property: p106 forms part of
Connector Node: Yes
Connected to: Admin model
Data type: Resource-instance-list
Name of concepts collection: N/A
Cardinality: 0..n
Necessity: Mandatory if applicable

Scope note (description of node)
The function of this node is to capture the relationship between the different administrative levels and uses the E41 class with the p106 property. The relationship structure indicates which level 2 are part of which level 1 and which level 1 are part of level 0. This allows for the capture of the complete administrative hierarchy for each administrative entry.

Example:
Bulawayo

Protocol for entry:

- Determine the hierarchical position of the administrative area being recorded and connect it to the name of the large administrative unit of which it forms part.
- For level 0 this node will remain empty

Node Name: Country
CIDOC CRM Class: E41 Appellation
CIDOC CRM Property: p1 is identified by
Connector Node: No
Connected to: N/A
Data type: String
Name of concepts collection: N/A
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)

This node captures the name of the country in which the administrative area being recorded is located in and uses the E41 class with the p1 property. A core function of this node is to allow for more effective search functionality within Arches.

Example:
Botswana
Senegal

Protocol for entry:

- Add the name of the country in which the administrative area is located using the corresponding name for the country in the HUM dataset.

Semantic Node: Administrative area level

Node Name: Administrative level
CIDOC CRM Class: E55 Type
CIDOC CRM Property: p150 defines typical parts of
Connector Node: No
Connected to: N/A
Data type: Controlled-list
Name of concepts collection: Administrative level
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)
This function of the node is to record what administrative level the relevant administrative area belongs to and uses the E55 class with the p95 property. The resource model records administrative level 0 to 2. Administrative levels are the nested geographic units governments and data systems use to organise territory — from the country level down to local neighbourhoods — and are commonly labelled ADM0, ADM1, ADM2, etc. The ADM is an internationally used shorthand indicating each nested level of the administrative system. Each country can have a different name that is used when referencing a specific level, which necessitates a system that can be universally applied. In this system, ADM0 is the uppermost level and reflects the boundaries of the country. ADM1 will be the level just below this, and ADM2 the level below that etc.

Example:
Level 0
Level 1

Protocol for entry:

- Select the required level from the controlled vocabulary that reflects the position of the entry within the level structure of the relevant country.

Semantic Node: Geometry

Node Name: Geometry (WKT)
CIDOC CRM Class: E53 Place
CIDOC CRM Property: L54 is same as
Connector Node: No
Connected to: N/A
Data type: String
Name of concepts collection: N/A
Cardinality: 1..n
Necessity: Still to be decided if it will be added

Scope note (description of node)
This node provides information related to the geographical extent of each entry as it relates to the earth surface. The node uses the E53 class with the L54 (CIDOC CRM 6.2 extension CRMdig v3.2.1) property. The geometry is based on the World Geodetic System (WGS) 84 system, with the longitude and latitude points being presented in Well-known Text (WKT) format. Within the Arches system, geometry is stored as a GeoJSON-feature-collection, with upload formats being Shapefile (SHP), Keyhole Markup Language (KML) and the Geometry JavaScript Object Notation (geoJSON) format. Once uploaded to Arches, all geometry is transformed into geoJSON format.

Example
POLYGON ((-7 25.5,-6.75 25.5,-6.75 25.25,-7 25.25,-7 25.5))

Protocol for entry:

- All geometry values in the row must be in WKT format using the WGS 84.

Node Name: Description
CIDOC CRM Class: E62 String
CIDOC CRM Property: p3 has note
Connector Node: No
Connected to: N/A
Data type: String
Name of concepts collection: N/A
Cardinality: 0..n
Necessity: If applicable

Scope note (description of node)
This is a general text node that uses the E62 class with the p3 property. It describes the relevant entry.

Example:

Protocol for entry:

- This is a general text node
- Special characters should be avoided if possible, including: , " ' % # : ;

Node Name: Comment
CIDOC CRM Class: E62 String
CIDOC CRM Property: P3 has note
Connector Node: No
Connected to: N/A
Data type: String
Name of concepts collection: N/A
Cardinality: 0..n
Necessity: If applicable

Scope note (description of node)

This is a general, all-purpose node that captures any information that is regarded as required for the specific entry but is not captured in any of the other nodes present within the resource model. It uses the E62 String with the L54 (CIDOC CRM 6.2 extension CRMdig v3.2.1) property.

Example
Remember to avoid special characters as far as possible.

Protocol for entry:
- This is a general text node
- Special characters should be avoided if possible, including: , " ' % # : ;

Semantic Node: Arches upload details

Node Name: Uploader name
CIDOC CRM Class: E82 Actor Appellation
CIDOC CRM Property: p139 has alternative form
Connector Node: No
Connected to: N/A
Data type: String
Name of concepts collection: N/A
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)

The function of this node is to capture the information related to the name of the individual who uploaded the information of the associated entry to the Arches database, and uses the E82 class with the p139 prosperity.

Example:

Renier Hendrik van der Merwe

Protocol for entry:

- The name of the individual who uploaded the data to the Arches database should be the full version of the name.

Node Name: Upload date
CIDOC CRM Class: E63 Beginning of Existence
CIDOC CRM Property: p92 brought into existence
Connector Node: No
Connected to: N/A
Data type: EDTF
Name of concepts collection: N/A
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)
The function of this node is to capture the date on which the information of the associated entry was uploaded to the Arches database and uses the E62 class with the p92 property.

Example:
2026-02-19

Protocol for entry:

- The date should follow the Gregorian calendar
- Use year, month, and day separated by hyphens (e.g., YYYY-MM-DD)
- Any other format will cause an error during upload

Node Name: Resource last modified at
CIDOC CRM Class: E63 Beginning of Existence
CIDOC CRM Property: p92 brought into existence
Connector Node: No
Connected to: N/A
Data type: EDTF
Name of concepts collection: N/A
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)
The function of this node is to capture the date on which any modification was made to any nodes related to the associated entry. This node uses the E63 class with the p92 property.

Example:

2026-03-05

Protocol for entry:

- Every time any modification is made to a specific entry within the resource model, the date should be captured as a new value
- All dates already present should be kept and unmodified
- The date should follow the Gregorian calendar
- Use year, month, and day separated by hyphens (e.g., YYYY-MM-DD)
- Any other format will cause an error during upload

Section 3: Comments

Reference

Aalber, T.,Balze, D., Bekiari, C.,Boudouri, L., Crofts, N., Dunsire, G.,Eide, Ø., Gill, T.,Goerz, G.,Hagedorn-Saupe, M., Hiebel, G., Holmen, J., Inkari,J., Iorizzo, D., Kotipelt, J., Kraus, S., Lampe, K.H., Lamsfus, C., Lindenthal, J., Nyman, M., Riva, P., Rold, L., Smiraglia, R., Stein, R., Stiff, M and Žumer, M. 2015. *Definition of the CIDOC Conceptual Reference Mode*. https://cidoc-crm.org/sites/default/files/cidoc_crm_version_6.2.pdf. Access: 2026-03-12

Appendix A

Country	Code (ISO 3166)
Benin	BEN
Botswana	BWA
Democratic Republic of the Congo	COD
Ethiopia	ETH
Ghana	GHA
Guinea-Bissau	GNB
Kenya	KEN
Mali	MLI
Mozambique	MOZ
Nigeria	NGA
Senegal	SEN
Sudan	SDN
Tanzania	TZA
The Gambia	GMB
Togo	TGO
Zimbabwe	ZWE